

UNIVERSITY OF PRETORIA

Introduction to OrCAD

(EBN 111: Electricity and Electronics)



orcad

a Cadence product family

04 August 2017

Department of Electrical, Electronic and
Computer Engineering



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Overview



1. General Practical Admin
2. SPICE, PSpICE and OrCAD
3. Practical 1
4. Example Question Demo



General Practical Admin



- All practicals take place in the CAEC labs (on the top floor of ENG II)
- The practicals will take place during the weeks of:
 - Practical 1: 14 - 18 August
 - Practical 2: 04 - 08 September
 - Practical 3: 16 - 20 October
- Component Box handout
 - 28 – 01 August

General Practical Admin



- Practical sessions – See Clickup for the most up to date version.
- You **MUST** attend the practical in your allocated session.
 - Register will be taken before entry
 - You will NOT be allowed to participate in a different session
 - If you miss a practical session you will need to submit an OFFICIAL doctors note within 3 days.
- Missing a practical **WILL** result in **exam refusal**.
- If you require a change, fill in the Practical Session Change form on Clickup.

General Practical Admin



- Please take note of the laboratory rules, it is enforced strictly:
 - No food or liquids are allowed.
 - No caps or hats (including beanies) may be worn.
 - **Closed shoes must be worn at all times (pumps and similar semi-open shoes are also not allowed).**
 - Test rules apply (no talking outside of your specified groups)
- For a full list see:
 - <http://www.up.ac.za/en/eece/article/2008415/undergraduate-laboratory-rules>
- If you are still confused see FAQ in Practicals/Prakties folder on clickup

SPICE and OrCAD



- SPICE- Simulation Program with Integrated Circuit Emphasis
 - Circuit analysis tool
 - Program that solves circuit equations
 - Free software, but difficult to use

PSPICE and OrCAD

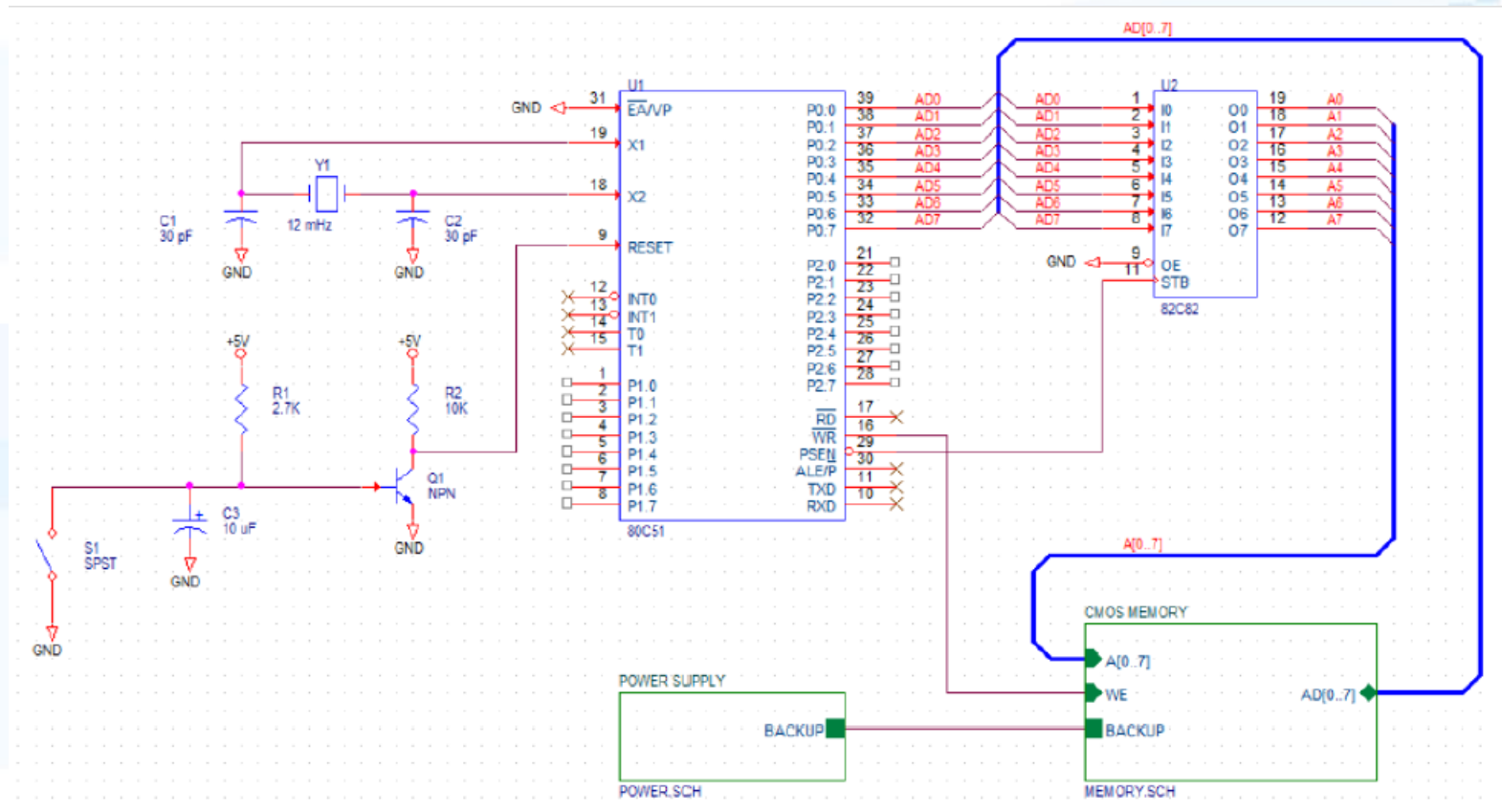


- Commercial implementation of SPICE
- Provides a GUI interface for circuit design and analysis
- Although many tools are available in the full OrCAD suite, we will only use OrCAD Capture (a schematic input tool) and the PSPICE components for analysis.

OrCAD Capture [Demo]



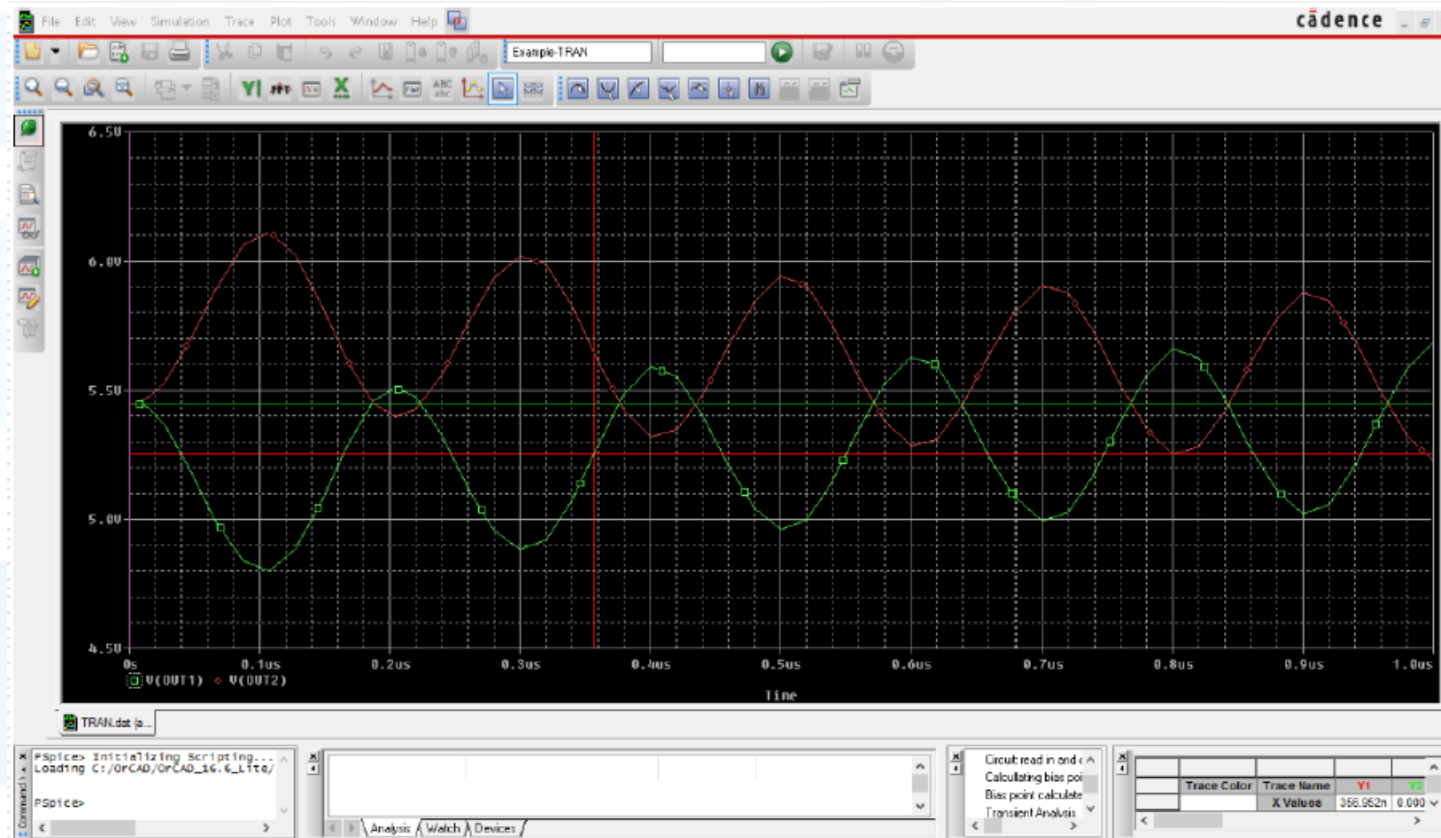
- Used to design circuits.



PSPICE [Demo]



- Simulate and analyze the designed circuit



PSPICE and OrCAD



- Essential tools for computer aided circuit analysis and design
- Reduces the time it takes to analyze and design complex circuits
 - Replaces complex calculation that would have been performed by hand with the press of a button, that gives near instant results
- These tools can be used in many subsequent modules. Even though the modules may not explicitly require its use, many students prefer to use it when complex designs are required

Practical 1



- Only OrCAD will be used in Practical 1
- You will start working with real components in practicals 2 and 3

Practical 1 – Steps to Complete NB!



- Before the practical:
 - 1, Complete the tutorial
 - 2, Complete the pre-practical assignment
- During the practical session (3 hour session):
 - 1, Complete the pre-practical test
 - 2, Complete the practical assignment/online evaluation
- These steps are summarized in the “*instructions.pdf*” file on Clickup.
 - (clickup>Practicals/Prakties>Practical 1>Practical 1 Instructions)

Practical 1



Tutorial

- How to get access to OrCAD
 - Download location
- How to use OrCAD
- Prepare you for the pre-practical assignment

Practical 1



Pre-practical Assignment

- Required for entry into the practical
- **Bring the completed assignment to the practical (No entry allowed without this)**
- Does not count for marks
- Preparation for the practical itself



Practical 1

Practical Test (20% of Practical 1 mark)

- Completed in groups of two
- 30 min
- **CLOSED** book
- The question tests your circuit knowledge up to now and the following Practical 1 material:
 - Practical Lecture Slides, Practical Tutorial, Pre-prac Assignment, EBN circuits up to now



Practical 1

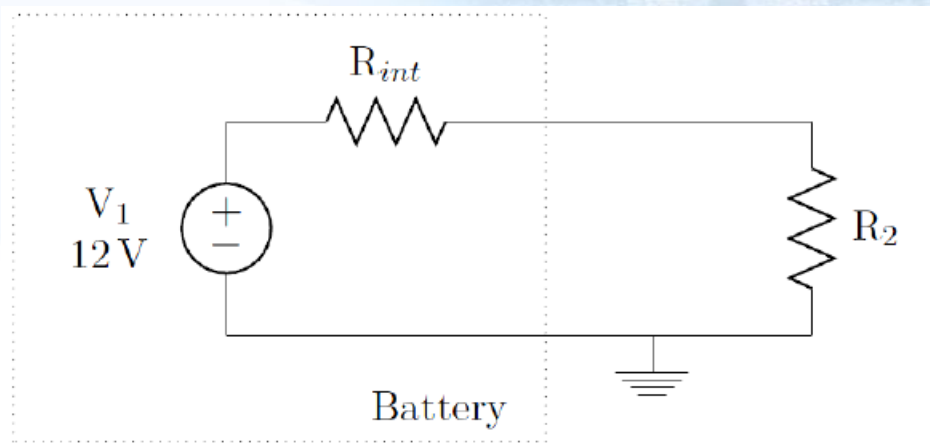
Practical Assignment (80% of Practical 1 mark)

- Completed in groups of two
- Includes some randomly generated individual questions
- Completed during the remaining 2 to 2,5 hours
- Clickup Test where you have to simulate a variety of circuits using OrCAD
- All practical material will be available to you during this time
- The demi's and AL's will also be available for help



Example Question Demo

- Consider the circuit below
 - The battery has an internal resistance of 5Ω
 - R_2 is the load resistance
- Complete a simulation of the circuit with $R_2 = 5\Omega$
 - Determine the current through R_2
 - Determine the voltage across R_2
- Sequentially set the value of V_1 to five values in the range 0 – 12V
 - Determine the load voltage (V_2) across R_2 and the current for each resistor



Conclusion



Remember

- Closed shoes must be worn at all times during the practical
- All steps that must be completed for the practical is included in the file “*instructions.pdf*”
- The documents referenced in this presentation will be published on Clickup later today

For any further enquiries please send an email to
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